

Multiple Choice

1. **Answer:** C

Options: A) ancient heavily cratered highlands; B) dark lavas inside volcanic calderas; C) dark lavas filling older impact basins; D) the bright regions on the Moon

Note: Correct option: C - dark lavas filling older impact basins

2. **Answer:** B

Options: A) To measure interstellar mass.; B) To measure galactic distances.; C) To measure galactic energy-density.; D) To measure interstellar density.

Note: Correct option: B - To measure galactic distances.

3. **Answer:** D

Options: A) the Sun mainly emits blue light.; B) the atmosphere absorbs mostly blue light.; C) molecules scatter red light more effectively than blue light.; D) molecules scatter blue light more effectively than red light.

Note: Correct option: D - molecules scatter blue light more effectively than red light.

4. **Answer:** C

Options: A) the entire planets are made mostly of metal.; B) metals condensed first in the solar nebula and the rocks then accreted around them.; C) metals sank to the center during a time when the interiors were molten throughout.; D) radioactivity created metals in the core from the decay of uranium.

Note: Correct option: C - metals sank to the center during a time when the interiors were molten throughout.

5. **Answer:** D

Options: A) Perseus; B) Cygnus; C) Scorpius; D) Leo

Note: Correct option: D - Leo

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Its distance from you'

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'The minimization of gravitational potential energy.'

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '380 to 740 nm.'

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'The period of rotation is the same on every point on the disk.'

Long Form Response

11. **Answer:** Laniakea. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** Distance at which one astronomical unit measures one arcsecond from Earth.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key